

EEE 491 Lab Assignment

Lab-DCT: Discrete Cosine Transform

1. Technical Specifications

Lab-DCT shall read energy data of each filter bank from Lab-MEL's dual-port RAM. Lab-DCT shall calculate discrete cosine transform of the energy data and the results shall be stored in a simple dual-port RAM. The RAM size shall be at least eight words. Lab-DCT output data width of the RAM shall be 16-bit. Use "Block Memory generator" and "Multipliers" from the Vivado "IP Catalog".

Lab-DCT in FPGA shall have the following input/output (IO) signals.

- "reset_in" input control signal, which resets all the registers in the design.
- "clock_in" input clock signal of the 100MHz clock oscillator on the Basys-3 board.
- "mel_addr_out" output signal that addresses the read-only port of the dual-port RAM in Lab-MEL.
- "mel_data_in" input signal from the read-only port of the dual-port RAM in Lab-MEL.
- "mem_addr_in" input signal that addresses the read-only port of the dual-port RAM in Lab-DCT.
- "mem_data_out" output signal of the read-only port of the dual-port RAM in Lab-DCT.
- "start_in" input control signal, which starts the DCT filtering function. This signal is active high asserted on a single clock pulse (refer to waveform in Lab-ADC assignment).
- "ready_out" output status signal that indicates if DCT filtering function is completed and output data are written to dual-port RAM of Lab-DCT and the function is ready for the next execution. This signal is active high and asserted low just after the start signal is asserted (refer to waveform in Lab-ADC assignment).

After verifying your Lab-DCT design by using your test-bench simulation results, integrate Lab-DCT with the Lab-PCB, ADC board, Lab-ADC, Lab-WINDOW, Lab-CTRL, Lab-FFT, Lab-MEL and Lab-DEBUG. Connect Lab-DEBUG to Lab-DCT output. Control and observe the process sequence using start_in signal of Lab-CTRL and ready_out signals of the components. On the Basys-3 board, use a button for start_in signaling and a LED for each ready_out signals of the components.

2. Demonstration

- Present your test bench simulation results: inputs, outputs (waveforms). Test bench generates a number of test data for band energies which may be stored in a ROM (filled with data in COE file), starts the Lab-DCT, and waits for the ready signal and then it reads data from the dual-port RAM of Lab-DCT. Validate the DCT function by the simulation output waveforms and by the results obtained on MATLAB.
- Configure your integrated design on Basys-3 board. Use start_in button and ready_out LEDs of the components to observe/verify process sequence. If you need you may add more LEDs for further observations.
- Prepare the Lab-CTRL demonstration set up. Run the integrated system and transfer the DCT output data of each frame to MATLAB. Compare and verify MEL filter results with the Lab-MATLAB results for sinus input frequencies: 100Hz, 500Hz, 1KHz, 4KHz with 1.5V peak-to-peak amplitude for each frequency.

3. Guidance

- You can refer to <https://link.springer.com/content/pdf/bbm:978-3-319-49220-9/1.pdf>
- You can use the function given below for DCT. $DCT(n)$ becomes cepstral coefficients where n is from 0 to $N-1$ and N may typically be between 7 and 13. $E(m)$ is the energy in the mel filter bank m . M is the total number of mel filter banks.

$$\text{DCT}(n) = \sum_{m=0}^{M-1} E(m) \cos\left(\frac{\pi n(m+0.5)}{M}\right)$$

- DCT function can be implemented by using a ROM. Generate Cosine values for n and m variables on MATLAB, include them in a ROM memory, and use them as the multiplication coefficient of the energy data. Represent Cosine data in 16-bit.